

GOVERNMENT OF INDIA — MARITIME ENVIRONMENTAL PROTECTION
DIVISION
HEADQUARTERS INDIAN COAST GUARD • NATIONAL TECHNICAL
RESEARCH ORGANISATION (NTRO)

TECHNICAL EVIDENCE DOSSIER & STATUTORY NOTICE OF VIOLATION
ISSUED UNDER SECTION 356E, MERCHANT SHIPPING ACT 1958 & MARPOL 73/78 ANNEX I

Dossier Ref:	ICG-MEP-DOSSIER-20260905-GREA	Date of Issue:	05 Sep 2026, 16:04 UTC
Maritime Sector:	Andaman & Nicobar Islands / Malacca Strait Transit	Legal Classification:	ILLEGAL DISCHARGE (CRIMINAL)
Target Vessel:	CHEM STAR III	IMO / MMSI:	9411234 / 357890123
Flag State:	Panama	Attribution Certainty:	81.7% (CONFIRMED)

1. SATELLITE RADAR REMOTE SENSING (SAR) OBSERVATION

Sensor Platform:	Sentinel-1A C-Band SAR	Acquisition Timestamp:	2026-09-04 11:45:00 UTC
Swath Mode & Pol:	Interferometric Wide (IW) Swath / VV	Ground Resolution:	10.0 meters/pixel
Observed Slick Centroid:	6.8385° N, 93.4007° E	Total Spill Surface Area:	36.34 km²
Estimated Discharge Mass:	799.5 Metric Tonnes	Lookalike Rejection Score:	65.0% Mineral Crude Confidence
ADIOS Weathering State:	Heavy Chocolate Mousse (Highly Viscous Emulsion)	Evaporative Loss / Mousse:	22.3% Evaporated / 75.0% Water Uptake

2. REVERSE LAGRANGIAN HYDRODYNAMIC DRIFT ANALYSIS (ORIGIN ATTRIBUTION)

Using an advection-diffusion Lagrangian numerical formulation integrated with real-time oceanographic current vector fields ($\Delta u = -0.32$ m/s, $\Delta v = 0.08$ m/s) and Ekman surface windage (3.2% Stokes drift factor with 15° Coriolis deflection), the slick plume trajectory was hindcasted backwards in time. The particle convergence analysis resolved the **exact discharge origin point**:

Spill Origin Point (x_0, y_0):	6.81041° N, 93.59886° E	Calculated Slick Age:	11.0 Hours prior to SAR pass
Estimated Discharge Time:	T - 11.0h (05 Sep 2026 approx 02:40 IST)	Total Hydrodynamic Drift:	20.14 km displacement

3. AIS CORRELATION & KINEMATIC BEHAVIORAL ANOMALIES

Analysis of **2 vessels** operating in the sector filtered down to **2 corridor candidates**. Vessel **CHEM STAR III** exhibited definitive spatio-temporal intersection with the discharge epicenter:

Evaluation Metric	Observed Telemetry Value	Anomaly Assessment	Score Weight
Closest Point of Approach (CPA)	0.76 NM (1.40 km)	Exact spatial co-location with slick root	91.5/100
Temporal Coincidence	$\Delta t = 1.00$ hours from origin	Precise temporal synchronization at t_0	70.7/100
Speed Over Ground (SOG)	Drop: 14.4 → 5.2 kts	Severe speed drop (14.4 -> 5.2 kts) matching illegal bilge purge profile	100.0/100
Vessel Risk Category	Chemical Tanker (DWT: 52,000 T)	High capacity oily water / slop separator	90.0/100

4. NON-COOPERATIVE / DARK VESSEL RADAR SURVEILLANCE AUDIT

TACTICAL RADAR TARGET INTERCEPTION: CFAR point-target detection extracted **12 radar contacts** with metallic corner-reflector signatures ($>+5$ dB σ^0) that broadcast **ZERO AIS transponder telemetry**.

Radar Target ID	Position (Lat, Lon)	RCS (σ^0 dB) / Length	Threat Classification
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RADAR-TGT-627	6.7421° N, 93.4752° E	28.3 dB / ~361.7m	HIGH PRIORITY SUSPECT (COVERT DISCHARGE)
RADAR-TGT-2592	6.7812° N, 93.3178° E	28.8 dB / ~305.1m	UNREGISTERED MARITIME CONTACT
RADAR-TGT-2736	6.7839° N, 93.4408° E	29.2 dB / ~664.1m	HIGH PRIORITY SUSPECT (COVERT DISCHARGE)

5. COASTAL HAZARD & BEACHING INTERCEPTION ALERT

CRITICAL SHORELINE ALERT: Hydrodynamic forecasting indicates slick impact along the coastline within 18.5 hours. Interception coordinates: 6.9272° N, 93.1078° E. Immediate mobilization of ICG Pollution Response (PR) Vessels and containment booms is ordered.

5. STATUTORY DIRECTIVES & DETENTION ORDER

NOTICE TO MASTER, OWNER, AND FLAG ADMINISTRATION:
Pursuant to powers vested under Section 356J and Section 356E of the Merchant Shipping Act, 1958:

1. **PORT STATE CONTROL INTERCEPTION:** The Principal Officer, Mercantile Marine Department (MMD), is directed to **DETAIN** vessel **CHEM STAR III (IMO: 9411234)** upon entry into any Indian port or anchorage until mandatory oil record book (ORB Part I & II) inspection and bilge manifold swab sampling are concluded.

2. **FINANCIAL SECURITY:** A maritime environmental lien of **INR 25,00,00,000 (Twenty-Five Crore Rupees)** is hereby levied to cover offshore containment, dispersant spraying, and coastal remediation costs.

3. **FLAG STATE NOTIFICATION:** A copy of this satellite radar and AIS kinematic forensic package is dispatched to the International Maritime Organization (IMO) and the Maritime Administration of Panama.

Forensic Chain-of-Custody SHA-256 Hash:
f1c4a76d802f5333213338b09a1fbbb98ad0836503fd0f5c21206c4d5fbb7371

Authorized Signatory:
Inspector General (Operations & MEP)
Indian Coast Guard Headquarters, New Delhi